

The **Pima Indians Diabetes Dataset** contains only **768 observations** and 8 features. Because of its small scale, modern machine learning models can easily overfit it. [1, 2, 3, 4]

The following tabular overview highlights larger, high-quality alternative datasets for diabetes and health classification tasks, ranging from thousands to hundreds of thousands of records:

Larger Alternative Datasets

Dataset Name [5, 6, 7, 8, 9]	Approximate Size	Key Advantages	Primary Use Case
CDC BRFSS Diabetes Indicators	253,000+ rows	Massively scalable, zero missing values, includes lifestyle habits	Large-scale classification, lifestyle analysis
Diabetes 130-US Hospitals (UCI)	101,000+ rows	Clinical data, multi-class outcomes, 55 separate features	Hospital readmission, clinical tracking
Early Stage Diabetes Risk (UCI)	520 rows	Larger feature set (16 signs/symptoms) despite a smaller footprint	Symptom-based early risk triage
NHANES Diabetes Data	Varies (Thousands)	Lab results, physical exams, and demographic tracking	Population health, epidemiological studies

Comprehensive Breakdown of Alternatives

- **CDC BRFSS Diabetes Health Indicators (Kaggle)**
 - **Size:** ~253,680 instances with 21 features.
 - **Why it is better:** Gathered directly via the CDC's Behavioral Risk Factor Surveillance System. It moves beyond pure lab metrics to include behavioral risk factors like smoking, physical activity, diet, and healthcare access.
 - **Best for:** Building highly scalable binary classification models with robust demographics. [5, 10, 11, 12, 13]
- **Diabetes 130-US Hospitals Dataset (UCI Machine Learning Repository)**
 - **Size:** 101,766 hospital encounters across 10 years.
 - **Why it is better:** It offers 55 distinct attributes, focusing entirely on clinical care. It

includes laboratory test results, medications (like insulin changes), and patient demographics.

- **Best for:** Predicting hospital readmission risk (3-class outcome: No readmission, readmitted in <30 days, readmitted in >30 days) and evaluating medication adjustments. [6, 14, 15, 16, 17]
- **NHANES (National Health and Nutrition Examination Survey)**
 - **Size:** Thousands of patient records updated continuously in multi-year continuous cycles.
 - **Why it is better:** Administered by the US government, this dataset combines interviews, physical examinations, and actual laboratory medical testing data.
 - **Best for:** Advanced feature engineering, epidemiological research, and combining lab tests (A1C, fasting glucose) with lifestyle surveys. [8, 10, 18]
- **Early Stage Diabetes Risk Prediction Dataset (UCI)**
 - **Size:** 520 instances with 17 attributes.
 - **Why it is better:** While physically smaller in row count than Pima, it offers double the feature density. It focuses squarely on early clinical symptoms (such as polyuria, polydipsia, sudden weight loss, and visual blurring) rather than strictly tracking continuous blood chemistry values.
 - **Best for:** Triage pipelines and non-invasive questionnaire-based machine learning models. [5, 7, 19, 20, 21]

To help narrow down the selection, please let me know if you need a dataset that focuses primarily on **lifestyle/survey data** or one consisting of **strict clinical/laboratory records**.

- [1] <https://www.frontiersin.org>
- [2] <https://www.sciencedirect.com>
- [3] <https://www.sciencedirect.com>
- [4] <https://arxiv.org>
- [5] <https://arxiv.org>
- [6] <https://www.researchgate.net>
- [7] <https://pmc.ncbi.nlm.nih.gov>
- [8] <https://pmc.ncbi.nlm.nih.gov>
- [9] <https://pmc.ncbi.nlm.nih.gov>
- [10] <https://link.springer.com>
- [11] <https://arxiv.org>
- [12] <https://ieeexplore.ieee.org>
- [13] <https://arxiv.org>
- [14] <https://www.mdpi.com>
- [15] <https://www.sciencedirect.com>
- [16] <https://machinelearningmastery.com>
- [17] <https://arxiv.org>
- [18] <https://link.springer.com>
- [19] <https://blog.devgenius.io>
- [20] <https://www.mdpi.com>
- [21] <https://www.sciencedirect.com>